

VBF-C1T1R1 dfmea - Virtual Battery Factory deliverable

DFMEA - Champion Cell R4-V3-margin (c1_t1_r1)

VBF-C1T1R1-DFMEA-001 I Virtual Battery Factory

S/O/D = severity/occurrence/detection engineering assessments (1-10 scale);

RPN = S x O x D. Mitigation and verification columns cite measured evidence where available; no assessment below is a simulated value.

I Failure mode I Cause I Effect I S I O I D I RPN I Mitigation in design I Verification evidence I

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I Lithium plating during 4C charge I anode-side concentration/kinetic polarization at 20 A I capacit

I Cell overheat > 60 C during 4C I I2R + entropic heating at 20 A I degradation, safety margin loss

I Thermal runaway under overcharge to 4.7 V I excessive overcharge reactions I fire/venting I 10 I 1

I Electrolyte salt depletion at high current I low transference / diffusivity I concentration polari

I Re-plating from over-cooling I h too high slows kinetics I plating despite cooling I 6 I 2 I 2 I 2

I Thin separator (8 um) mechanical failure I manufacturing defect, dendrite I internal short I 9 I 2

I Thin current collectors (8/6 um) fatigue I tab weld stress I contact loss I 4 I 2 I 3 I 24 I stand

Top residual risks: thin-separator mechanical integrity (RPN 54) and salt depletion (RPN 36) - both mitigated but recommended for cell-level manufacturing validation.